

# MIT integration bee - 2019

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## Notación y Consideraciones Generales

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A lo largo del documento vamos a utilizar la siguiente notación, en algunos problemas más complejos utilizaremos la notación usual:

$$\{ f(x) \}_a^b := \int_a^b f(x) dx, \quad [f(x)] := \frac{d}{dx} f(x)$$

Recursos obtenidos de: <https://math.mit.edu/~yyao1/integrationbee.html>

Herramientas que te pueden ser útiles:

- Calculadora gráfica: <https://www.desmos.com/calculator?lang=es>
- Calculadora de integrales: <https://mathdf.com/es/>
- Calculadora de integrales: <https://www.wolframalpha.com/>
- Motor de búsqueda para fórmulas en latex <https://approach0.xyz/search/>
- Te ayuda con ideas: <https://chat.deepseek.com>

## Ejercicio 1

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$$\int_0^{2\pi} \tan(\cos(x)) dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \{ \tan(\cos(x)) \}_0^\pi + \{ \tan(\cos(x)) \}_\pi^{2\pi} \\ &= \{ \tan(\cos(x)) \}_0^\pi + \{ \tan(-\cos(x)) \}_0^\pi \quad (1) \\ &= \{ \tan(\cos(x)) \}_0^\pi - \{ \tan(\cos(x)) \}_0^\pi \quad (2) \\ &= 0 \end{aligned}$$

- (1):  $\cos(x + \pi) = \cos(x)\cos(\pi) - \sin(x)\sin(\pi) = -\cos(x)$
- (2):  $\tan(x)$  es una función impar

## Ejercicio 2

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$$\int \frac{x+1}{x(x+\log x)} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{1+1/x}{x+\ln x} \right\} \\ &= \ln|x+\ln(x)| + C \quad (1) \end{aligned}$$

- (1):  $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

## Ejercicio 3

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$$\int e^{x+e^x} + e^{x-e^x} dx$$

— — — Demostración — — —

$$I = \{ e^u + e^{-u} \} \quad (1)$$

$$= 2 \{ \cosh(u) \} \quad (2)$$

$$= 2 \sinh(u) + C \quad (3)$$

$$\bullet (1): \begin{cases} u &= e^x \\ du &= e^x dx \end{cases}$$

$$\bullet (2): \cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\bullet (3): [\sinh(x)]' = \cosh(x) \vee [\cosh(x)]' = \sinh(x)$$

## Ejercicio 4

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$$\int_{-1/2}^{1/2} \frac{dx}{1-x^2}$$

--- Demostración ---

$$\begin{aligned} I &= \frac{1}{2} \left\{ \frac{1}{1-x} + \frac{1}{1+x} \right\}_{-1/2}^{1/2} \\ &= \frac{1}{2} ( -\ln|1-x| + \ln|1+x| )_{-1/2}^{1/2} \\ &= \ln(3) \end{aligned}$$

## Ejercicio 5

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$$\int_0^2 2^{\log x} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ 2^{\frac{\log_2(x)}{\log_2(e)}} \right\}_0^2 \quad (1) \\ &= \left\{ 2^{\frac{\log_2(x)}{\log_2(e)}} \right\}_0^2 \\ &= \left\{ x^{\frac{1}{\log_2(e)}} \right\}_0^2 \\ &= \left\{ x^{\ln(2)} \right\}_0^2 \quad (1) \\ &= \left( \frac{x^{\ln(2)+1}}{\ln(2)+1} \right)_0^2 \\ &= \frac{2^{\ln(2)+1}}{\ln(2)+1} \end{aligned}$$

$$\bullet (1): \log_a(x) = \frac{\log_b(x)}{\log_b(a)}$$

## Ejercicio 6

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$$\int_{-2\pi}^{2\pi} (\cos 3x + \sin 2x)(-\sin 2019x + \cos 3x) dx$$

--- Demostración ---

$$\begin{aligned}
I &= \left\{ -\cos(3x) \sin(2019x) + \cos^2(3x) - \sin(2x) \sin(2019x) + \sin(2x) \cos(3x) \right\}_{-2\pi}^{2\pi} \\
&= -\left\{ \cos(3x) \sin(2019x) \right\}_{-2\pi}^{2\pi} + \left\{ \sin(2x) \cos(3x) \right\}_{-2\pi}^{2\pi} + \left\{ \cos^2(3x) - \sin(2x) \sin(2019x) \right\}_{-2\pi}^{2\pi} \\
&= \left\{ \cos^2(3x) - \sin(2x) \sin(2019x) \right\}_{-2\pi}^{2\pi} \\
&= 2 \left\{ \cos^2(3x) - \sin(2x) \sin(2019x) \right\}_0^{2\pi}
\end{aligned} \tag{1}$$

$$\begin{aligned}
I_1 &= \left\{ \cos^2(3x) \right\}_0^{2\pi} \\
&= \frac{1}{2} \left\{ \cos(6x) + 1 \right\}_0^{2\pi} \\
&= \frac{1}{2} \left( \frac{\sin(6x)}{6} + x \right)_0^{2\pi} \\
&= \pi
\end{aligned} \tag{3}$$

$$\begin{aligned}
I_2 &= \left\{ \sin(2x) \sin(2019x) \right\}_0^{2\pi} \\
&= \left\{ \sin(2x) \sin(2019x) \right\}_0^{2\pi} \\
&= \frac{1}{2} \left\{ \cos(2017) - \cos(2021) \right\}_0^{2\pi} \\
&= \frac{1}{2} \left( \frac{\sin(2017x)}{2017} - \frac{\sin(2021x)}{2021} \right)_0^{2\pi} \\
&= 0
\end{aligned} \tag{4}$$

$$I = 2\pi$$

- (1): Funciones impares en intervalos simétricos
- (2): Funciones pares en intervalos simétricos
- (3):  $\cos(2x) = 2 \cos^2(x) - 1$
- (4):  $\cos(a - b) - \cos(a + b) = 2 \sin(a) \sin(b)$

## Ejercicio 7

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$$\int \cos(x) \cdot \cos(\sin x) \cdot \cos(\sin(\sin x)) \, dx$$

— — — Demostración — — —

$$I = \left\{ \cos(u) \cos(\sin(u)) \right\} \tag{1}$$

$$= \left\{ \cos(t) \right\} \tag{2}$$

$$= \sin(t) + C$$

$$\bullet (1): \begin{cases} u &= \sin(x) \\ du &= \cos(x) \, dx \end{cases}$$

$$\bullet (2): \begin{cases} t &= \sin(u) \\ dt &= \cos(u) \, du \end{cases}$$

## Ejercicio 8

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$$\int_0^\infty \frac{e^{-\frac{2019}{4t^2}}}{t^2} \, dt$$

— — — Demostración — — —

$$\begin{aligned}
I &= \left\{ \frac{e^{-\frac{a}{t^2}}}{t^2} \right\}_0^\infty \\
&= \left\{ e^{-\alpha x^2} \right\}_0^\infty
\end{aligned} \tag{1}$$

$$\begin{aligned}
&= \left\{ e^{-(\sqrt{\alpha}x)^2} \right\}_0^\infty \\
&= \frac{1}{\sqrt{\alpha}} \left\{ e^{-u^2} \right\}_0^\infty
\end{aligned} \tag{2}$$

$$= \frac{1}{2} \sqrt{\frac{\pi}{\alpha}} \tag{3}$$

$$= \sqrt{\frac{\pi}{2019}}$$

- (1):  $\begin{cases} x &= \frac{1}{t} \\ dx &= -\frac{1}{t^2} dt \end{cases}$
- (2):  $\begin{cases} u &= \sqrt{\alpha}x \\ du &= \sqrt{\alpha} dx \end{cases}$
- (3): Integral Gaussiana  $\{e^{-x^2}\}_{-\infty}^{\infty} = \sqrt{\pi}$

## Ejercicio 9

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$$\int \sin(\sqrt{x}) dx$$

--- Demostración ---

$$I = 2 \{ t \sin(t) \} \quad (1)$$

$$= 2 ( -t \cos(t) + \sin(t) ) + C \quad (2)$$

- (1):  $\begin{cases} x &= t^2 \\ dx &= 2t dt \end{cases}$

- (2):  $\left| \begin{array}{c|c} D & I \\ t & \sin(t) \\ 1 & -\cos(t) \\ 0 & -\sin(t) \end{array} \right|$

## Ejercicio 10

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$$\int_0^1 \frac{\sqrt{x}}{1+x} dx$$

--- Demostración ---

$$\begin{aligned} I &= 2 \left\{ \frac{t^2}{1+t^2} \right\}_0^1 \quad (1) \\ &= 2 \left\{ 1 - \frac{1}{1+t^2} \right\}_0^1 \\ &= 2 ( t - \arctan(t) )_0^1 \\ &= 2 ( 1 - \pi/4 ) \end{aligned}$$

- (1):  $\begin{cases} x &= t^2 \\ dx &= 2t dt \end{cases}$

## Ejercicio 11

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$$\int_0^{2\pi} \cos(x) \cos(2x) \cos(3x) dx$$

--- Demostración ---

$$I = \frac{1}{2} \{ \cos(x)(\cos(5x) + \cos(x)) \}_0^{2\pi} \quad (1)$$

$$= \frac{1}{2} \{ \cos(x) \cos(5x) + \cos^2(x) \}_0^{2\pi}$$

$$= \frac{1}{2} \left\{ \frac{\cos(6x) + \cos(4x)}{2} + \cos^2(x) \right\}_0^{2\pi} \quad (1)$$

$$= \frac{1}{4} \{ \cos(6x) + \cos(4x) \}_0^{2\pi} + \frac{1}{2} \{ \cos^2(x) \}_0^{2\pi}$$

$$= \frac{1}{2} \{ \cos^2(x) \}_0^{2\pi} \quad (2)$$

$$= \frac{1}{4} \{ \cos(2x) + 1 \}_0^{2\pi} \quad (3)$$

$$= \frac{1}{4} \{ \cos(2x) \}_0^{2\pi} + \frac{1}{4} \{ 1 \}_0^{2\pi}$$

$$= \frac{1}{4} \{ 1 \}_0^{2\pi} \quad (2)$$

$$= \frac{\pi}{2}$$

- (1):  $\cos(a+b) + \cos(a-b) = 2\cos(a)\cos(b)$
- (2):  $\cos(x+\pi) = -\cos(x)$
- (3):  $\cos(2x) = 2\cos^2(x) - 1$

## Ejercicio 12

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$$\lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} e^{-x^{2n}} dx$$

— — — Demostración — — —

Considere la siguiente sucesión de funciones  $f_n(x) = e^{-x^{2n}}$  donde  $x \in \mathbb{R}$ .

Necesitamos que la sucesión  $\{f_n(x)\}_{n \geq 1}$  converga uniformemente a una función  $f(x)$  para todo  $x \in \mathbb{R}$ , para poder aplicar el siguiente teorema: [https://en.wikipedia.org/wiki/Uniform\\_convergence](https://en.wikipedia.org/wiki/Uniform_convergence) (nos permite conmutar la integral por el límite).

Suponiendo que tenemos esta convergencia uniforme, entonces:

$$\begin{aligned} \lim_{n \rightarrow \infty} I_n &= \lim_{n \rightarrow \infty} \{f_n(x)\}_{-\infty}^{\infty} \\ &= \lim_{n \rightarrow \infty} 2 \{f_n(x)\}_0^{\infty} & (1) \\ &= 2 \lim_{n \rightarrow \infty} (\{f_n(x)\}_0^1 + \{f_n(x)\}_1^{\infty}) \\ &= 2 \left( \left\{ \lim_{n \rightarrow \infty} f_n(x) \right\}_0^1 + \left\{ \lim_{n \rightarrow \infty} f_n(x) \right\}_1^{\infty} \right) \\ &= 2 \left( \{1\}_0^1 + \left\{ \lim_{n \rightarrow \infty} f_n(x) \right\}_1^{\infty} \right) & (2) \\ &= 2 \left( \{1\}_0^1 + \{0\}_1^{\infty} \right) & (3) \\ &= 2 \end{aligned}$$

- (1): Función par en intervalo simétrico
- (2): Si  $x \in (0, 1)$ , entonces  $x^n < x < 1$  y por lo tanto  $\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} e^{-x^{2n}} = e^0 = 1$
- (3): Si  $x \in (1, \infty)$ , entonces  $1 < x < x^n$  y por lo tanto  $\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} e^{-x^{2n}} = 0$

**Nota:**

1. Convergencia uniforme:

- [https://en.wikipedia.org/wiki/Uniform\\_convergence](https://en.wikipedia.org/wiki/Uniform_convergence)
- <https://alephsub0.org/material-nuevo/jonathan-ortiz/intercambiar-la-integral-y-el-limite/>
- (video que demuestra el teorema para convergencia uniforme)  
<https://www.youtube.com/watch?v=SPq6-kEs9CU>

## Ejercicio 13

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$$\int_0^e x^{\frac{1}{\log x}} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \{e\}_0^e & (1) \\ &= e(x)_0^e \\ &= e^2 \end{aligned}$$

- (1):  $x^{\frac{1}{\ln(x)}} = e^{\ln(x^{\frac{1}{\ln(x)}})} = e$

## Ejercicio 14

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$$\int_0^{\pi/100} \frac{\sin(20x) + \sin(19x)}{\cos(20x) + \cos(19x)} dx$$

--- Demostración ---

$$I = \left\{ \frac{2 \sin\left(\frac{39}{2}x\right) \cos\left(\frac{x}{2}\right)}{\cos(20x) + \cos(19x)} \right\}_0^{\pi/100} \quad (1)$$

$$= \left\{ \frac{2 \sin\left(\frac{39}{2}x\right) \cos\left(\frac{x}{2}\right)}{2 \cos\left(\frac{39}{2}x\right) \cos\left(\frac{x}{2}\right)} \right\}_0^{\pi/100} \quad (2)$$

$$= \left\{ \tan\left(\frac{39}{2}x\right) \right\}_0^{\pi/100}$$

$$= -\frac{2}{39} \left( \ln \left| \cos\left(\frac{39}{2}x\right) \right| \right)_0^{\pi/100}$$

$$= -\frac{2}{39} \ln \left| \cos\left(\frac{39\pi}{200}\right) \right|$$

- (1): Considere el siguiente desarrollo:

$$\begin{aligned} 2 \sin\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right) &= 2 (\sin(a/2) \cos(b/2) + \cos(a/2) \sin(b/2)) (\cos(a/2) \cos(b/2) + \sin(a/2) \sin(b/2)) \\ &= 2 (\sin(a/2) \cos(b/2) + \cos(a/2) \sin(b/2)) (\cos(a/2) \cos(b/2) + \sin(a/2) \sin(b/2)) \\ &= 2 (\sin(a/2) \cos(a/2) + \sin(b/2) \cos(b/2)) \\ &= \sin(a) + \sin(b) \end{aligned}$$

- (2): Considere el siguiente desarrollo:

$$\begin{aligned} 2 \cos\left(\frac{a+b}{2}\right) \cos\left(\frac{a-b}{2}\right) &= 2 (\cos(a/2) \cos(b/2) - \sin(a/2) \sin(b/2)) (\cos(a/2) \cos(b/2) + \sin(a/2) \sin(b/2)) \\ &= 2 (\cos^2(a/2) \cos^2(b/2) - \sin^2(a/2) \sin^2(b/2)) \\ &= 2 (\cos^2(a/2) \cos^2(b/2) - (1 - \cos^2(a/2))(1 - \cos^2(b/2))) \\ &= 2 (\cos^2(a/2) + \cos^2(b/2) - 1) \\ &= \cos(a) + \cos(b) \end{aligned}$$

## Ejercicio 15

$$\int (e^x \cos^2(x) + e^x \cos(x) \sin(x) - e^x \sin^2(x)) dx$$

--- Demostración ---

$$I = \left\{ e^x (\cos^2(x) + \cos(x) \sin(x) - \sin^2(x)) \right\} \quad (1)$$

$$= \left\{ e^x (\cos(2x) + \cos(x) \sin(x)) \right\} \quad (2)$$

$$= \frac{\sin(2x)}{2} e^x + C \quad (3)$$

- (1):  $\cos(2x) = \cos^2(x) - \sin^2(x)$

- (2):  $\sin(2x) = 2 \sin(x) \cos(x)$

- (3):  $[e^x f(x)] = e^x (f(x) + f'(x))$

## Ejercicio 16

$$\int_0^{\pi/2} \frac{\sin x}{\sin(x + \pi/4)} dx$$

--- Demostración ---

$$I = \sqrt{2} \left\{ \frac{\sin(x)}{\sin(x) + \cos(x)} \right\}_0^{\pi/2} \quad (1)$$

$$= \sqrt{2} \left\{ \frac{\cos(\theta)}{\sin(x) + \cos(x)} \right\}_0^{\pi/2} \quad (2)$$

$$\begin{aligned} 2I &= \sqrt{2} \left\{ \frac{\sin(\theta)}{\sin(x) + \cos(x)} + \frac{\cos(\theta)}{\sin(x) + \cos(x)} \right\}_0^{\pi/2} \\ &= \sqrt{2} \{ 1 \}_0^{\pi/2} \\ &= \frac{\pi}{\sqrt{2}} \end{aligned}$$

$$I = \frac{\pi}{2\sqrt{2}}$$

- (1):  $\sin(x + \pi/4) = \frac{1}{\sqrt{2}}(\sin(x) + \cos(x))$

- (2):  $\begin{cases} x &= \pi/2 - \theta \\ dx &= -d\theta \end{cases}$

## Ejercicio 17

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$$\int \frac{dx}{x + \sqrt[3]{x}}$$

--- Demostración ---

$$I = 3 \left\{ \frac{t}{t^2 + 1} \right\} \quad (1)$$

$$= \frac{3}{2} \ln |t^2 + 1| + C \quad (2)$$

- (1):  $\begin{cases} x &= t^3 \\ dx &= 3t^2 dt \end{cases}$

- (2):  $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

## Ejercicio 18

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$$\int_0^2 x^{x^2+1} (2 \log(x) + 1) dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ x^{x^2} (2x \ln(x) + x) \right\}_0^2 \\ &= \left( x^{x^2} \right)_0^2 \end{aligned} \quad (1)$$

$$= 2^{2^2} - 1 \quad (2)$$

$$= 15 \quad (2)$$

- (1):  $[x^{x^2}] = [e^{x^2 \ln(x)}] = e^{x^2 \ln(x)} (2x \ln(x) + x) = x^{x^2} (2x \ln(x) + x)$

- (2):  $\lim_{x \rightarrow 0} x^{x^2} = \lim_{x \rightarrow 0} e^{x^2 \ln(x)} = e^{\lim_{x \rightarrow 0} x^2 \ln(x)} = e^0 = 1$  (aplicando L'hospital)

## Ejercicio 19

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$$\int \frac{2x^3 - 1}{x(x^3 + 1)} dx$$

--- Demostración ---

$$\begin{aligned}
I &= \left\{ \frac{2(x^3+1)-3}{x(x^3+1)} \right\} \\
&= \left\{ \frac{2}{x} - \frac{3}{x(x^3+1)} \right\} \\
&= 2 \ln|x| - 3 \left\{ \frac{1}{x(x^3+1)} \right\} \\
&= 2 \ln|x| - 3 \left\{ \frac{1}{x(x^3+1)} \right\} \\
I_1 &= \left\{ \frac{1}{x(x^3+1)} \right\} \\
&= - \left\{ \frac{u^2}{u^3+1} \right\} \quad (1) \\
&= -\frac{1}{3} \left\{ \frac{3u^2}{u^3+1} \right\} \\
&= -\frac{1}{3} \ln|u^3+1| + C \quad (2) \\
&= -\frac{1}{3} \ln \left| \left( \frac{1}{x} \right)^3 + 1 \right| + C
\end{aligned}$$

$$\begin{aligned}
I &= 2 \ln|x| + \ln \left| \left( \frac{1}{x} \right)^3 + 1 \right| + C \\
&= \ln|x^3+1| - \ln|x| + C
\end{aligned}$$

$$\bullet (1): \begin{cases} x &= \frac{1}{u} \\ dx &= -\frac{1}{u^2} du \end{cases}$$

$$\bullet (2): [\ln(f(x))] = \frac{f'(x)}{f(x)}$$

## Ejercicio 20

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$$\int \cos(\arctan(x)) dx$$

--- Demostración ---

$$\begin{aligned}
I &= \{ \cos(u) \sec^2(u) \} \quad (1), (2) \\
&= \{ \sec(u) \} \\
&= \ln|\sec(u) + \tan(u)| + C \quad (3), (4) \\
&= \ln|\sqrt{x^2+1} + x| + C
\end{aligned}$$

$$\bullet (1): \begin{cases} u &= \arctan(x) \\ du &= \frac{1}{1+x^2} dx \end{cases}$$

$$\bullet (2): 1 + \tan^2(x) = \sec^2(x)$$

$$\bullet (3): \sec(x) = \frac{\sec(x)(\sec(x)+\tan(x))}{\sec(x)+\tan(x)}$$

$$\bullet (4): [\sec(x) + \tan(x)] = \sec(x)(\sec(x) + \tan(x))$$